KKR & KSR INSTITUTE OF TECHNOLOGY & SCIENCES
(AUTONOMOUS)

DEPARTMENT OF INFORMATION TECHNOLOGY

B.Tech – II Year I Semester

S.No.	Category	Title	L	T	P	C	IM	EM	TM
1	BS&H	Discrete Mathematics & Graph Theory	3	0	0	3	30	70	100
2	PC	Object Oriented Programming Though JAVA	3	0	0	3	30	70	100
3	ES	Digital Logic & Computer Organization	3	0	0	3	30	70	100
4	PC	Advanced Data Structures & Algorithm Analysis	3	0	0	3	30	70	100
5	BS&H	Universal Human Values – Understanding Harmony	2	0	0	3	30	70	100
6	PC	Advanced Data Structures & Algorithm Analysis Lab	0	0	3	1.5	30	70	100
7	PC	Object Oriented Programming Though JAVA Lab	0	0	3	1.5	30	70	100
8	Skill EC	Basic Web Development	0	0	2	2	-	50	50
9	AC	Environmental Science	2	0	0	-	-	-	-
Total			**16**	**0**	**8**	**20**	**210**	**540**	**750**

B.Te]ch – II Year II Semester

S. No.	Category	Title	L	T	P	C	IM	EM	TM
1	Management Course	Optimization Techniques	2	0	0	2	30	70	100
2	ES/ BS	Probability & Statistics	3	0	0	3	30	70	100
3	PC	Operating Systems	3	0	0	3	30	70	100
4	PC	Database Management Systems	3	0	0	3	30	70	100
5	PC	Python Programming	3	0	0	3	30	70	100
6	PC	Database Management Systems Lab	0	0	3	1.5	30	70	100
7	PC	Python Programming Lab	0	0	3	1.5	30	70	100
8	Skill EC	Full Stack Development	0	0	2	2	-	50	50
9	BS&H	Design Thinking & Innovation	2	0	0	1	-	50	50
Total			**16**	**0**	**8**	**20**	**210**	**590**	**800**

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II Year II Semester	FULL STACK DEVELOPMENT (SKILL ENHANCEMENT COURSE-2) (Bootstrap, REACT JS, NODE JS)	L	T	P	C
		0	1	2	2

Course Objectives:

The main objectives of the course are:

- To implement the static web pages using HTML and do client-side validation using JavaScript.
- To design and work with databases using Java.
- To develop an end-to-end application using java full stack.
- To introduce Node JS implementation for server-side programming.
- To experiment with single page application development using React.

Course Outcomes:

At the end of the course, the student will be able to

CO 1: build a custom website with HTML, CSS, and Bootstrap and little JavaScript.

CO 2: demonstrate Advanced features of JavaScript and learn about JDBC.

CO 3: develop Server – side implementation using Java technologies like.

CO 4: develop the server – side implementation using Node JS.

CO 5: design a Single Page Application using React.

List of Experiments:

[HTML, CSS, Bootstrap, Java Script]

1. Build a responsive web application for shopping cart with registration, login, catalog and cart pages using CSS3 features, flex and grid.
2. Make the above web application responsive web application using Bootstrap framework.
3. Use JavaScript for doing client – side validation of the pages implemented in experiment 1 and experiment 2.

[React JS]

4. Install ReactJS and Environment Setup.
5. Creating a simple ReactJS Application.
6. Event Handling in ReactJS.
7. Use Props in React JS to Customize React Components.
8. Passing Data Between React JS Components Using Props.
9. Create a TODO application in react with necessary components and deploy it into github.
10. Create a service in react that fetches the weather information from openweathermap.org and the display the current and historical weather information using graphical representation using chart.js

11, Create a react application for the student management system having registration, login, contact, about pages and implement routing to navigate through these pages.

[Node JS]

Install NodeJS and Environment Setup.

12. Create a calculator Module
13. File operations using nodeJS
14. Connect to MySQL database and perform database operations.

15. Parse URL parameters, Parse a JSON file.
16. Create a custom server using http module and explore the other modules of Node JS like OS, path, event.

REFERENCE BOOKS:

1. Jon Duckett, Beginning HTML, XHTML, CSS, and JavaScript, Wrox Publications, 2010
2. Bryan Basham, Kathy Sierra and Bert Bates, Head First Servlets and JSP, O'Reilly Media, 2nd Edition, 2008.
3. Vasan Subramanian, Pro MERN Stack, Full Stack Web App Development with Mongo, Express, React, and Node, 2 nd Edition, A Press.

ONLINE RESOURCES:

1. https://www.almabetter.com/bytes/tutorials/reactjs/higher-order-components
2. https://www.tutorialkart.com/nodejs/node-js-examples/#gsc.tab=0